Collecting crash dumps for Desktop Support

Article KB0010050 | Owner: Desktop Support | Subcategory: Blue screen / crash | Applies to: Windows 11 23H2/24H2 on Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Managed via Microsoft Intune, Users experiencing a STOP error or random system crash

Summary

This article guides users and Desktop Support agents on how to collect Windows crash dump files (minidump or kernel memory dump) after a blue-screen (BSOD) event on Acme-managed laptops. The dump enables deeper analysis of faulty drivers, updates, or hardware issues. Follow the self-help steps to capture the dump and attach it to your ticket; agents will then analyse the file and determine the appropriate fix.

Symptoms

- STOP: 0x000000EF (CRITICAL_PROCESS_DIED) or similar bugcheck code displayed on a blue screen
- System restarts automatically and shows a brief "Windows recovered from a serious error" message
- Event Viewer shows a Kernel-Power error (Event ID 41) with BugCheckCode parameter
- No obvious application error; crash occurs after Windows Update, driver install, or during heavy GPU usage

Common causes

- Faulty or incompatible graphics, network, or storage driver
- Recent Windows quality or feature update that introduced a bug
- Defective RAM or other hardware causing memory corruption
- Third-party software that hooks into kernel (e.g., antivirus, virtualization)
- Overheating or power-supply instability leading to spontaneous resets

Overview

When Windows encounters a fatal error it writes a crash dump to the local disk. By default Acme laptops are configured for a Small memory dump (256?KB) in %SystemRoot%\Minidump. Collecting this file gives Desktop Support the exact bugcheck code, faulting module, and stack trace needed to pinpoint the cause. The process below works for all supported Windows 11 builds and does not require admin rights beyond those already granted by Intune.

Prerequisites

Ensure the laptop is powered on and can boot to Windows (even if it crashes again later). You need access to the File Explorer and the ability to save files to your OneDrive or a USB flash drive. If the system repeatedly restarts before you can log in, try booting into Safe Mode (hold Shift while clicking Restart) to reach the desktop.

User self-help procedure: collect a crash dump

Follow these steps to locate the most recent dump file and attach it to your ticket. The steps assume the default Small memory dump setting; if your device was previously changed to a Kernel dump, the same folder will contain a larger .dmp file.

1. Open File Explorer and navigate to C:\Windows\Minidump. If you do not see the folder, type %SystemRoot%\Minidump in the address bar and press Enter.
2. Look for files with the .dmp extension. The most recent dump will have the latest date/time stamp. Right-click the file and choose Copy.
3. Paste the copied file into a location you can easily upload, such as your OneDrive folder (OneDrive\Acme\CrashDumps) or a USB flash drive.
4. If you cannot find any .dmp files, verify that dump collection is enabled: open Settings -> System -> About -> Advanced system settings -> Startup and Recovery -> Settings. Under 'Write debugging information' ensure 'Small memory dump (256?KB)' is selected and the dump directory is %SystemRoot%\Minidump. Click OK to apply.
5. Return to the IT self-service portal (https://it.acme.example) or open a new chat in the Teams channel "IT Help". Create or update your ticket, attach the copied .dmp file, and add a brief description of when the crash occurred (e.g., "BSOD after installing graphics driver version 31.0.15.4679 on 2024-09-12").
6. Submit the ticket. Desktop Support will be notified and will begin analysis.

Service-desk agent procedure

Once a ticket arrives with an attached dump file, follow the analysis workflow to identify the root cause and determine the appropriate remediation.

1. Download the attached .dmp file from the ticket or Teams chat to a secure analysis workstation (ensure the workstation has the Windows Debugging Tools installed - WinDbg preview from the Microsoft Store).
2. Open WinDbg, choose File -> Open Crash Dump, and select the .dmp file. Wait for the symbol download; Acme's symbols are pulled automatically from the Microsoft Symbol Server via Intune configuration.
3. Run the command !analyze -v to get a verbose bugcheck analysis. Note the BugCheckCode, the probable faulting module (MODULE_NAME), and the stack trace.
4. If the faulting module points to a driver (e.g., nvlddmkm.exe, igdkmd64.sys), check the driver version in Device Manager and compare against the approved baseline in the Driver Approval Matrix. Consider rolling back to the previous known-good version or installing the latest approved update from the Company Portal.
5. If the analysis indicates memory corruption (e.g., PAGE_FAULT_IN_NONPAGED_AREA with a random address), run Windows Memory Diagnostic or advise the user to schedule a full MemTest86+ pass via the Desktop Support hardware queue.
6. If no clear driver is implicated, review recent Windows Update KB numbers shown in the dump's 'Loaded module list' and cross-reference with known problematic updates in the KBs-Known-Issues wiki. Recommend uninstalling the update via Settings -> Windows Update -> Update history -> Uninstall updates.
7. Document the findings in the ticket, apply the remediation (driver rollback, update removal, RAM replacement, etc.), and request the user to test for stability over 24?hours.
8. If the crash persists after initial steps, escalate per the escalation guidance below.

Verification and closure

After applying a fix, ask the user to confirm that the system remains stable. Request they use the laptop normally for at least one full workday and monitor for any further BSODs. If no recurrence occurs, set the ticket status to Resolved and add a closure note summarizing the root cause and the action taken. Encourage the user to enable automatic crash-dump collection (already default) for future incidents.

When to escalate

If the initial analysis does not reveal a clear driver or update cause, or if the system continues to crash after driver rollback, memory test, and update removal, escalate to the Desktop Support senior team (Desktop Support L2) and/or the Hardware Engineering group. Include in the escalation: the full !analyze -v output, the list of loaded modules with timestamps, the BIOS/UEFI version, and any relevant Event Viewer logs (System and Application) covering the 24-hour window before each crash. Also note whether the device is under warranty and if any hardware diagnostics (Dell BIOS diagnostics, Lenovo Vantage, HP PC Hardware Diagnostics) have already been run.

Keywords: BSOD, blue screen, crash dump, minidump, kernel memory dump, WinDbg, !analyze -v, memory dump, STOP error, Driver rollback, Windows Update removal, MemTest86+, Hardware diagnostics, Intune, Company Portal, IT Help Teams, self-service portal